

## TEPLY KLYUCH, Russian Federation

WMO: 247710

Lat: 62.789N

Lon: 136.853E

Elev: 947

StdP: 14.20

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD<br>to 99.6% DB |           | WSF        |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|--------------------------|-----------|------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                          |           |            |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                     | PCWD      |            |
| (a)<br>1      | (b)<br>-60.6 | (c)<br>-57.6 | (d)<br>-64.7                  | (e)<br>0.1 | (f)<br>-60.1 | (g)<br>-61.9 | (h)<br>0.1 | (i)<br>-56.8 | (j)<br>6.1            | (k)<br>-24.1 | (l)<br>5.5 | (m)<br>-29.9 | (n)<br>2.0               | (o)<br>70 | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>22.9            | (c)<br>83.3     | (d)<br>63.6 | (e)<br>79.6 | (f)<br>61.4 | (g)<br>76.1 | (h)<br>60.6 | (i)<br>65.9         | (j)<br>77.6 | (k)<br>64.2 | (l)<br>75.4 | (m)<br>62.4 | (n)<br>72.6 | (o)<br>5.7           | (p)<br>270 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.1                            | 86.3 | 68.4 | 60.2 | 80.8 | 67.4 | 58.2 | 74.9 | 66.5 | 31.3          | 77.1 | 29.9 | 75.4 | 28.6 | 72.3 | 75.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 11.8          | (b) 9.7 | (c) 8.2 | (e) -63.4                  | (f) 87.7 | (g) 2.3            | (h) 1.7 | (i) -65.1                                          | (j) 88.9 | (k) -66.4  | (l) 89.9 | (m) -67.7  | (n) 90.8 | (o) -69.3  | (p) 92.1 |  |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         |                            |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.9        | -41.8 | -31.2 | -6.9  | 19.6  | 41.7  | 57.3  | 61.9  | 55.9  | 40.4  | 14.5  | -21.5 | -37.9 |
|                                                                        | DBStd       | 39.05       | 10.24 | 9.81  | 14.48 | 11.13 | 8.55  | 6.30  | 6.21  | 6.41  | 8.21  | 12.56 | 14.79 | 13.14 |
|                                                                        | HDD50       | 14380       | 2847  | 2274  | 1764  | 913   | 275   | 13    | 3     | 20    | 301   | 1101  | 2144  | 2725  |
|                                                                        | HDD65       | 19066       | 3312  | 2694  | 2229  | 1363  | 721   | 238   | 134   | 286   | 739   | 1566  | 2594  | 3190  |
|                                                                        | CDD50       | 838         | 0     | 0     | 0     | 0     | 19    | 232   | 371   | 204   | 12    | 0     | 0     | 0     |
|                                                                        | CDD65       | 51          | 0     | 0     | 0     | 0     | 0     | 8     | 37    | 6     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1153        | 0     | 0     | 0     | 0     | 6     | 307   | 678   | 161   | 1     | 0     | 0     | 0     |
|                                                                        | CDH80       | 255         | 0     | 0     | 0     | 0     | 0     | 57    | 175   | 23    | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 3.4         | 2.0   | 2.3   | 3.2   | 4.2   | 5.0   | 4.7   | 4.1   | 3.8   | 3.7   | 3.4   | 2.6   | 2.3   |
| (15) Precipitation                                                     | PrecAvg     | 14.6        | 0.4   | 0.2   | 0.3   | 0.6   | 1.5   | 2.2   | 2.9   | 2.2   | 1.9   | 1.3   | 0.7   | 0.3   |
|                                                                        | PrecMax     | 19.5        | 0.8   | 0.7   | 1.1   | 1.9   | 2.8   | 4.1   | 7.1   | 3.7   | 3.6   | 2.4   | 2.1   | 0.9   |
|                                                                        | PrecMin     | 11.5        | 0.1   | 0.1   | 0.1   | 0.2   | 0.6   | 1.1   | 1.1   | 1.1   | 0.4   | 0.4   | 0.2   | 0.0   |
|                                                                        | PrecStd     | 2.0         | 0.2   | 0.2   | 0.2   | 0.4   | 0.5   | 0.8   | 1.3   | 0.7   | 0.8   | 0.5   | 0.4   | 0.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | -1.4  | 6.3   | 33.3  | 49.5  | 73.5  | 85.3  | 88.3  | 82.9  | 70.2  | 44.5  | 31.2  | 14.0  |
|                                                                        |             | MCWB        | -2.1  | 5.4   | 27.0  | 38.3  | 52.1  | 63.3  | 65.7  | 64.4  | 54.6  | 37.6  | 30.2  | 13.4  |
|                                                                        | 2%          | DB          | -10.4 | -1.7  | 29.0  | 45.3  | 67.9  | 81.5  | 84.9  | 78.9  | 64.8  | 38.2  | 17.4  | 1.4   |
|                                                                        |             | MCWB        | -10.7 | -2.5  | 24.5  | 35.2  | 49.2  | 59.7  | 64.6  | 61.9  | 51.6  | 34.4  | 16.5  | 1.1   |
|                                                                        | 5%          | DB          | -19.3 | -8.9  | 25.3  | 41.7  | 63.5  | 77.7  | 81.5  | 74.8  | 60.6  | 34.7  | 8.4   | -9.7  |
|                                                                        |             | MCWB        | -19.6 | -9.6  | 21.2  | 32.8  | 47.5  | 58.2  | 63.6  | 60.8  | 49.5  | 32.1  | 7.5   | -10.1 |
|                                                                        | 10%         | DB          | -26.7 | -15.0 | 20.4  | 38.3  | 58.6  | 73.9  | 77.7  | 71.1  | 55.4  | 31.8  | 0.1   | -19.4 |
|                                                                        |             | MCWB        | -26.9 | -15.5 | 17.1  | 30.7  | 45.4  | 57.1  | 62.7  | 59.0  | 46.4  | 30.0  | -0.6  | -19.7 |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | -1.6  | 5.3   | 28.8  | 38.6  | 53.4  | 65.3  | 70.1  | 66.5  | 55.6  | 39.0  | 30.3  | 13.2  |
|                                                                        |             | MCDB        | -1.0  | 6.3   | 31.4  | 49.0  | 72.1  | 79.8  | 80.3  | 76.5  | 68.4  | 42.4  | 31.3  | 14.4  |
|                                                                        | 2%          | WB          | -10.6 | -2.3  | 24.5  | 35.6  | 50.1  | 62.9  | 66.8  | 64.6  | 53.1  | 35.3  | 16.6  | 0.5   |
|                                                                        |             | MCDB        | -10.2 | -1.6  | 29.1  | 43.4  | 65.4  | 76.8  | 79.0  | 74.0  | 61.5  | 37.5  | 17.4  | 0.9   |
|                                                                        | 5%          | WB          | -19.5 | -9.5  | 21.1  | 33.5  | 48.3  | 60.7  | 65.4  | 62.5  | 50.7  | 32.4  | 7.6   | -10.3 |
|                                                                        |             | MCDB        | -19.2 | -8.8  | 25.2  | 40.7  | 61.9  | 73.3  | 77.0  | 71.6  | 58.1  | 34.1  | 8.4   | -9.8  |
|                                                                        | 10%         | WB          | -26.9 | -15.4 | 16.9  | 31.2  | 46.2  | 59.0  | 63.7  | 60.5  | 47.8  | 29.6  | -0.3  | -19.8 |
|                                                                        |             | MCDB        | -26.7 | -14.9 | 20.5  | 37.6  | 56.0  | 70.4  | 74.7  | 68.8  | 53.9  | 31.9  | 0.3   | -19.6 |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 15.0  | 26.3  | 34.9  | 28.4  | 21.1  | 24.5  | 22.9  | 23.6  | 20.7  | 17.7  | 17.7  | 12.9  |
|                                                                        |             | MCDBR       | 22.2  | 31.1  | 32.2  | 25.9  | 31.4  | 34.5  | 31.7  | 30.8  | 30.2  | 18.7  | 21.5  | 23.4  |
|                                                                        | 5% WB       | MCWBR       | 21.9  | 29.8  | 26.9  | 17.2  | 16.1  | 15.7  | 13.9  | 15.6  | 17.1  | 14.1  | 20.5  | 22.2  |
|                                                                        |             | MCWBR       | 22.1  | 30.9  | 30.5  | 23.4  | 28.4  | 27.1  | 25.8  | 25.1  | 25.4  | 16.0  | 21.5  | 22.4  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.188 | 0.243 | 0.294 | 0.365 | 0.414 | 0.405 | 0.490 | 0.422 | 0.337 | 0.270 | 0.197 | 0.166 |
|                                                                        |             | taud        | 1.979 | 2.113 | 2.070 | 1.962 | 2.097 | 2.285 | 2.023 | 2.201 | 2.407 | 2.246 | 1.971 | 1.901 |
|                                                                        | Ebn at Noon |             | 153   | 226   | 252   | 253   | 252   | 259   | 228   | 236   | 241   | 212   | 139   | 91    |
|                                                                        |             | Edh at Noon | 10    | 21    | 33    | 45    | 44    | 37    | 47    | 36    | 24    | 18    | 10    | 5     |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 97    | 406   | 994   | 1622  | 1656  | 1724  | 1613  | 1262  | 711   | 375   | 150   | 43    |
|                                                                        | RadStd      |             | 6     | 17    | 38    | 64    | 101   | 160   | 125   | 108   | 99    | 34    | 11    | 2     |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page